

# Parvez Akhtar

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## PROFESSIONAL SUMMARY

Senior GenAI and backend engineer with 5+ years building scalable, event-driven distributed systems and production-grade Agentic AI workflows in Python (FastAPI). Integrates, scales, and deploys LLM applications with LangChain and LlamaIndex across AWS and Azure, containerized with Kubernetes and Docker and shipped through CI/CD. Reduced inference costs by ~90% for production extraction and billing engines while taking experimental AI to reliable, cost-efficient enterprise production.

## TECHNICAL SKILLS

**Languages:** Python (OOP, asynchronous programming/asyncio, concurrency, design patterns, memory management, performance optimization), Java, TypeScript, SQL.

**GenAI & Agentic AI:** LangChain, LlamaIndex, LangGraph, OpenAI Agents SDK, Anthropic SDK, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool/Function Calling, Structured Outputs, Agent Memory & Human-in-the-Loop, Guardrails & Prompt-Injection Defense, RAG (Retrieval-Augmented Generation) with Hybrid Search & Re-ranking, Fine-tuning (LoRA/QLoRA, Hugging Face), Prompt Engineering, Embeddings, Vector DBs (Qdrant/PGVector).

**Backend & APIs:** FastAPI, Django, Spring Boot, REST APIs, Event-Driven Architecture, Microservices, Serverless, Kafka.

**Cloud & DevOps:** AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, Kubernetes, Docker, Terraform, CI/CD (GitHub Actions), Temporal.io.

**Databases:** PostgreSQL, PGVector, MongoDB, Redis, Elasticsearch.

**Software Engineering:** SOLID, Clean Architecture, Design Patterns, Data Structures & Algorithms, Concurrency (Multithreading/Multiprocessing), Performance Optimization.

**ML & Data Engineering:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Snowflake Cortex, Apache Airflow, ETL Pipelines.

**LLMOps & Observability:** LLM Tracing & Eval (Langfuse, OpenTelemetry, RAGAS), SigNoz, ELK Stack, Cloud-Watch, Prometheus.

## PROFESSIONAL EXPERIENCE

### Bigstep Technologies

Remote

*Senior AI/ML Engineer, Lead Architect*

04/2025 – Present

- As **Lead Architect**, drove cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous business goals into production backend architectures.
- **Agentic AI & FinOps:** Orchestrated a serverless, **event-driven multi-agent** framework for Salesforce automation with **tool/function calling** over **MCP (Model Context Protocol)** and swarm-based orchestration. A pure **pay-per-execution** model cut idle compute, saving ~\$45k/year.
- **Client, Outcomelabs.ai (GenAI Platform):** Architected a scalable **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **"Strategy Router"** that classifies each field by complexity and routes ~60% to deterministic regex/rule-based parsers, reserving the LLM for the rest, cutting token spend ~40% with no measurable accuracy loss.
- **Client, VTS (current), Lease Abstraction:** Built an agentic **document-intelligence** pipeline that abstracts structured terms from commercial real-estate leases with **structured-output validation**. Diagnosed a **10-point accuracy regression** and redesigned prompt strategy and image preprocessing to recover the full drop, restoring field-level extraction to its ~**88% baseline** and unblocking downstream automation.
- **Search & Ranking:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5 (Normalized Discounted Cumulative Gain)** by ~15 points (**0.71 to 0.86**).
- **Core Engineering:** Constructed an entity matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing fine-tuning strategies for cost efficiency.

### KuttI (Incubated at IIM Lucknow)

Remote

*Founding Engineer*

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the entire codebase and infrastructure, ensuring a **zero-downtime handover** to the incoming engineering team.
- Engineered a distributed, **event-driven** document parsing pipeline using **FastAPI** and **Temporal.io**, processing 50k+ daily messages. Enforced resilient retry logic and dead-letter queues to guarantee no data loss.

- **Fine-tuned Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference costs by **90%** and latency by 300ms.
- Operationalized the ML lifecycle with **CI/CD evaluation gates (Ragas)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments below threshold and catching quality regressions before production.
- Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** under high-concurrency bursts.

#### **Amplicom**

Bengaluru, India

*SDE-2*

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to **Spring Boot microservices**, applying **SOLID and clean-architecture** principles to enable independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC (Role-Based Access Control)** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

#### **Amazon**

Bengaluru, India

*Software Development Engineer (SDE)*

03/2021 – 03/2023

- Owned the full SDLC for high-scale **microservices**, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of **ECS** services to use **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and a seamless customer transition.

#### **Publicis Sapient**

Remote, India

*Associate Data Engineer*

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

### EDUCATION

**National Institute of Technology Delhi**  
*B.Tech in Computer Science & Engineering*

Delhi, India  
2016 – 2020

### ACHIEVEMENTS

**Amazon "On the Hero Road" Award:** Recognized for exceptional impact on team delivery and engineering excellence.

**Udacity Computer Vision Nanodegree:** Full scholarship recipient (sponsored by Facebook).

**Deep Learning Specialization:** Completed 5-course specialization (Coursera) with perfect score.